

NestJS Server – API Quickstart

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Purpose

Compact, RAG-ready walkthrough to stand up a production-grade NestJS HTTP API.

1) Install & Scaffold

```
npm i -g @nestjs/cli
nest new api
```

2) Core Files

main.ts enables CORS, pipes, interceptors, and starts the HTTP server.
app.module.ts aggregates feature modules.

3) Global Validation

Use ValidationPipe with whitelist:true, transform:true, forbidNonWhitelisted:true.

4) Typical Module

- Controller: HTTP routing and DTO validation.
- Service: business logic (stateless where possible).
- Entities/Models: TypeORM/Prisma schemas.
- Providers: helpers, repositories, caching, etc.

5) Persistence Options

- TypeORM (Relational)
- Prisma (Relational/PlanetScale/etc)
- Mongoose (MongoDB)

6) Observability

- Logger (e.g., PinoLogger)
- Request scoped tracing headers
- Health checks at /health

7) Security Essentials

- Helmet, rate limiting, and secure cookies (if applicable)
- JWT AuthGuard & RolesGuard
- Input validation on every route

8) Deployment

- Docker multi-stage builds
- Fastify adapter for higher throughput
- NODE_ENV=production & cache control